# Trigonometric ratios

## Core Concepts: Trigonometric ratios

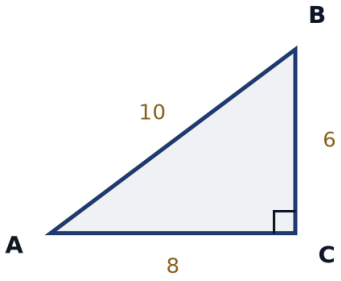
**1** For the model $f(x) = 3x + 5$ in trigonometric ratios, evaluate $f(2)$.

**a.** 11

**b.** 14

**c.** 6

**d.** 15




**2** Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

**a.** $x = 1$ and $x = 4$

**b.** $x = 1$ and $x = 3$

**c.** $x = -1$ and $x = -3$

**d.** $x = 4$ and $x = 3$

**3** A data set recorded for trigonometric ratios consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5

**d.** 21.5

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 3x + 5$ in trigonometric ratios, evaluate $f(3)$.

**a.** 14

**b.** 17

**c.** 9

**d.** 18

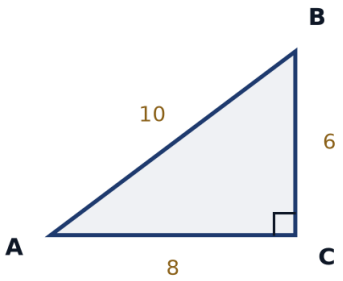
**5** Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ and $x = 4$

**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$




**6** A data set recorded for trigonometric ratios consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5

**d.** 21.5

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## Graphical & Geometric Interpretation

**7** For the model $f(x) = 3x + 5$ in trigonometric ratios, evaluate $f(4)$.

**a.** 17

**b.** 20

**c.** 12

**d.** 21

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**8** Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

**a.** $x = 3$ and $x = 6$

**b.** $x = 3$ and $x = 5$

**c.** $x = -3$ and $x = -5$

**d.** $x = 8$ and $x = 15$

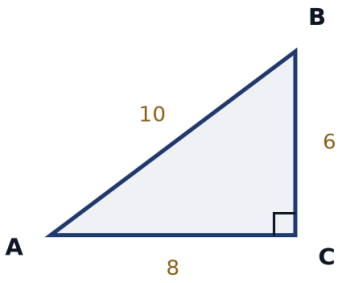
**9** A data set recorded for trigonometric ratios consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5

**d.** 21.5




## Exam-Style Problem Solving & Modeling

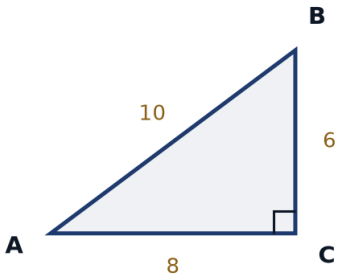
**10** For the model $f(x) = 3x + 5$ in trigonometric ratios, evaluate $f(5)$.

**a.** 20

**b.** 23

**c.** 15

**d.** 24




**11** Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

**a.** $x = 4$ and $x = 7$

**b.** $x = 4$ and $x = 6$

**c.** $x = -4$ and $x = -6$

**d.** $x = 10$ and $x = 24$

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**12** A data set recorded for trigonometric ratios consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** $17.5$

**b.** $18.5$

**c.** $19.5$

**d.** $21.5$

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